

Regularization Techniques

Omkar N

Contents

- ▶ Introduction
- ▶ What is Regularization?
- ▶ How does Regularization help reduce Overfitting?
- ▶ Different Regularization Techniques in Deep Learning
 - L2 & L1 regularization
 - Dropout
 - Data Augmentation
 - Early stopping

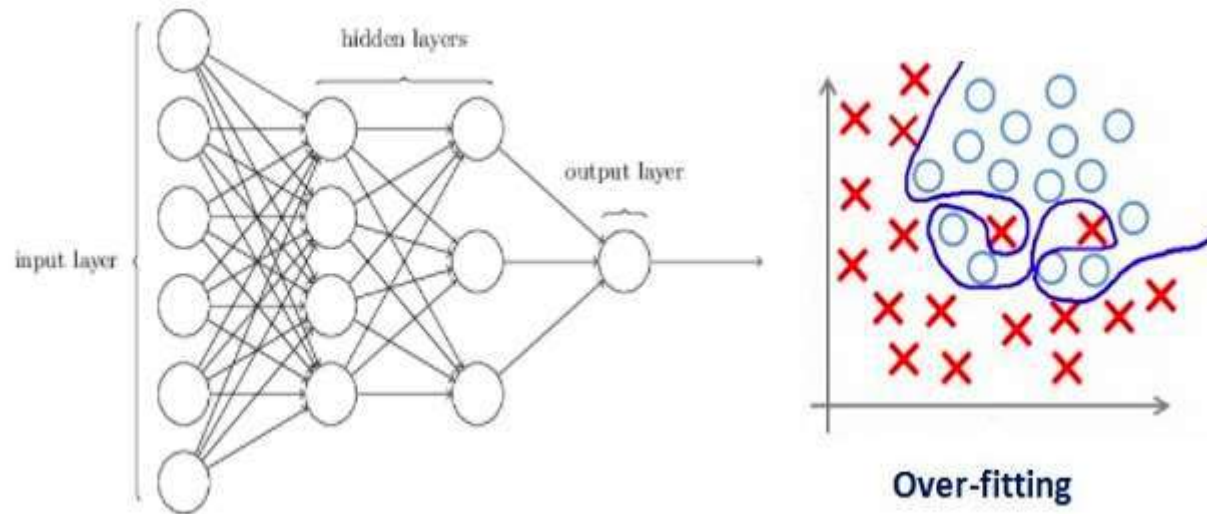
What is Regularization?

- ▶ Regularization is a technique used in machine learning and deep learning to prevent overfitting and improve the generalization performance of a model.
- ▶ It involves adding a penalty term to the loss function during training.
- ▶ This penalty discourages the model from becoming too complex or having large parameter values, which helps in controlling the model's ability to fit noise in the training data.
- ▶ Regularization methods include L1 and L2 regularization, dropout, early stopping, and more.
- ▶ By applying regularization, models become more robust and better at making accurate predictions on unseen data.



How does Regularization help reduce Overfitting?

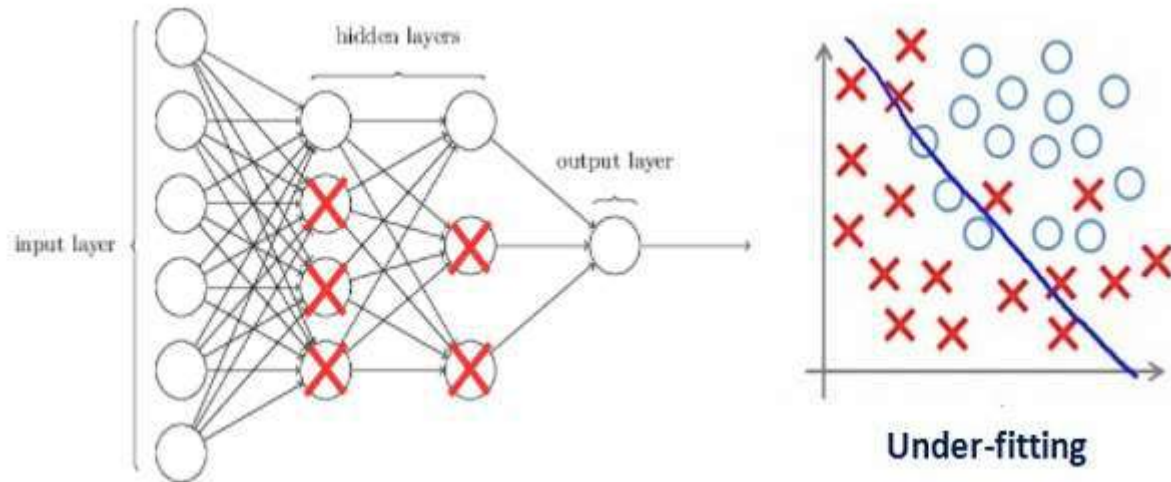
Let's consider a neural network which is overfitting on the training data as shown in the image below.



Contd..

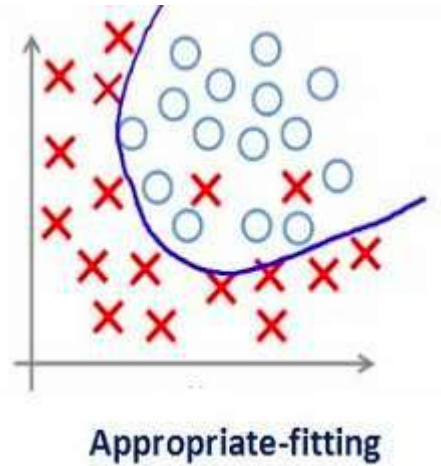
In deep learning, it actually penalizes the weight matrices of the nodes.

- Assume that our regularization coefficient is so high that some of the weight matrices are nearly equal to zero.



Contd..

- ▶ This will result in a much simpler linear network and slight underfitting of the training data.
- ▶ Such a large value of the regularization coefficient is not that useful. We need to optimize the value of regularization coefficient in order to obtain a well-fitted model as shown in the image below



L2 & L1 regularization

- ▶ L1 and L2 are the most common types of regularization. These update the general cost function by adding another term known as the regularization term.
- ▶ *Cost function = Loss (say, binary cross entropy) + Regularization term*

A regression model that uses L1 regularization technique is called **Lasso Regression** and model which uses L2 is called **Ridge Regression**.

- ▶ *The key difference between these two is the penalty term.*
- ▶ **Ridge regression** adds “squared magnitude” of coefficient as penalty term to the loss function. Here the *highlighted* part represents L2 regularization element

L2- Regularization

$$\sum_{i=1}^n (y_i - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

L1-Regularization

$$\sum_{i=1}^n (Y_i - \sum_{j=1}^p X_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

Contd..

Here, if *lambda* is zero then you can imagine we get back OLS. However, if *lambda* is very large then it will add too much weight and it will lead to under-fitting. Having said that it's important how *lambda* is chosen. This technique works very well to avoid over-fitting issue.

Again, if *lambda* is zero then we will get back OLS whereas very large value will make coefficients zero hence it will under-fit

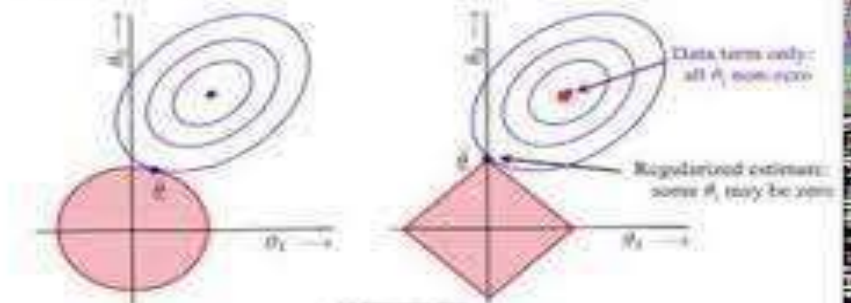
The key difference between these techniques is that Lasso shrinks the less important feature's coefficient to zero thus, removing some feature altogether. So, this works well for **feature selection** in case we have a huge number of features.

$$\sum_{i=1}^n (y_i - \sum_{j=1}^p x_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p \beta_j^2$$

$$\sum_{i=1}^n (Y_i - \sum_{j=1}^p X_{ij} \beta_j)^2 + \lambda \sum_{j=1}^p |\beta_j|$$

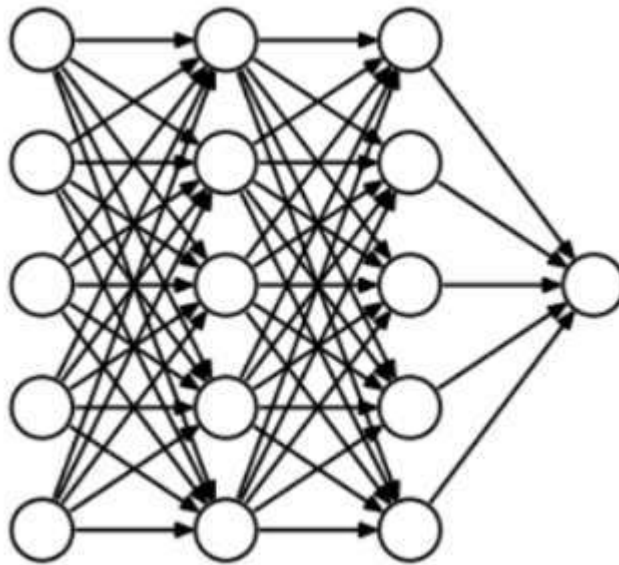
Regularization: L1 vs L2

- Estimate balances data term & regularization term
- Lasso tends to generate sparser solutions than a quadratic regularizer.



Dropout

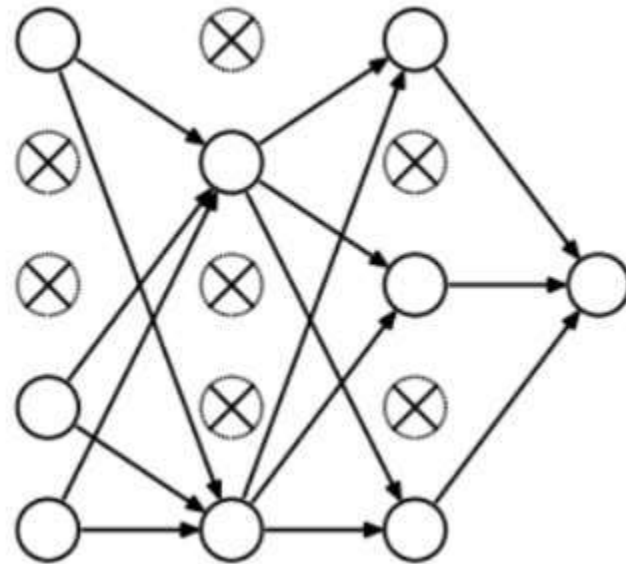
- ▶ This is the one of the most interesting types of regularization techniques. It also produces very good results and is consequently the most frequently used regularization technique in the field of deep learning.
- ▶ To understand dropout, let's say our neural network structure is akin to the one shown by side



Contd..

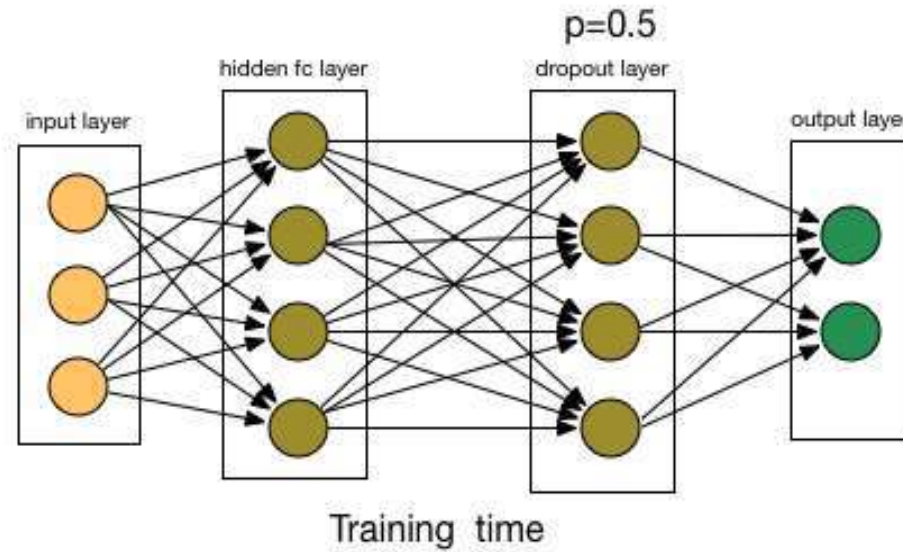
So what does dropout do? At every iteration, it randomly selects some nodes and removes them along with all of their incoming and outgoing connections as shown below.

- ▶ So each iteration has a different set of nodes and this results in a different set of outputs. **It can also be thought of as an ensemble technique in machine learning.**
- ▶ Ensemble models usually perform better than a single model as they capture more randomness. Similarly, dropout also performs better than a normal neural network model.



Contd..

This probability of choosing how many nodes should be dropped is the hyperparameter of the dropout function. As seen in the image above, dropout can be applied to both the hidden layers as well as the input layers.



Data Augmentation

- ▶ The simplest way to reduce overfitting is to increase the size of the training data. In machine learning, we were not able to increase the size of training data as the labeled data was too costly.
- ▶ But, now let's consider we are dealing with images. In this case, there are a few ways of increasing the size of the training data – rotating the image, flipping, scaling, shifting, etc. In the below image, some transformation has been done on the handwritten digits dataset.



This technique is known as data augmentation. This usually provides a big leap in improving the accuracy of the model. It can be considered as a mandatory trick in order to improve our predictions.

In keras, we can perform all of these transformations using ImageDataGenerator. It has a big list of arguments which you you can use to pre-process your training data.

Early stopping

Early stopping is a kind of cross-validation strategy where we keep one part of the training set as the validation set. When we see that the performance on the validation set is getting worse, we immediately stop the training on the model. This is known as early stopping.

